

Architecture & delivery visual guide

Trading Desk v3 / Architecture, workflows, state ownership and phase gates

Research first. Deterministic risk. Human authority.
Paper-only remains a valid outcome through Phase 5.

02 Domains and shared controls

04 From approval to economic effect

06 Four deterministic agent roles

08 Phases 0–3: establish evidence

10 State ownership and memory

12 LangGraph workflow

14 Phase outputs and admission gates

03 From evidence to approval

05 Implemented portfolio and simulator

07 Frozen models and research evidence

09 Phases 4–5: certify and operate

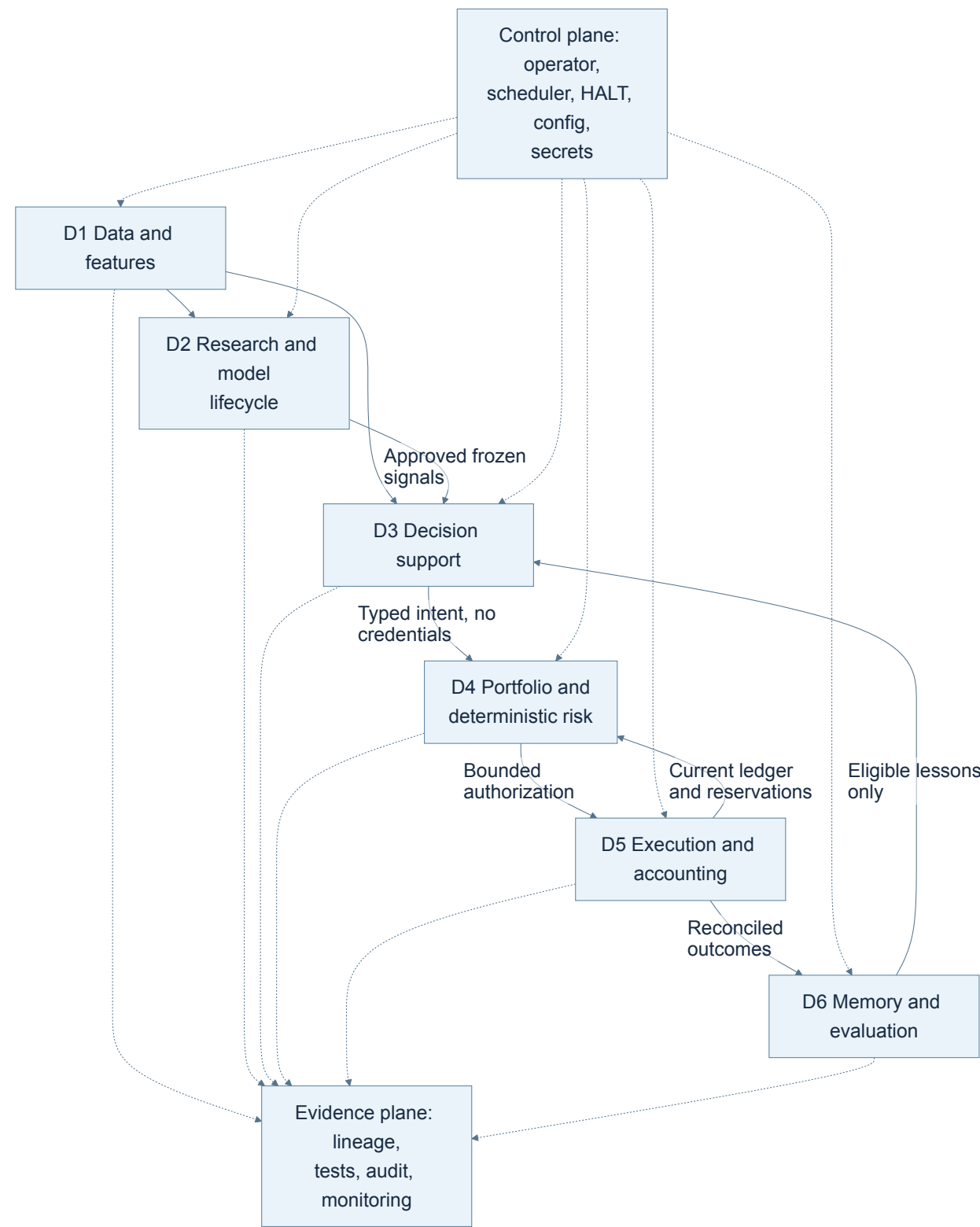
11 Verification and acceptance

13 Order lifecycle and uncertainty

Readiness: research and internal-paper prototype. Target diagrams describe the intended design, not implemented guarantees. No live broker adapter is provided.

Domains and shared controls

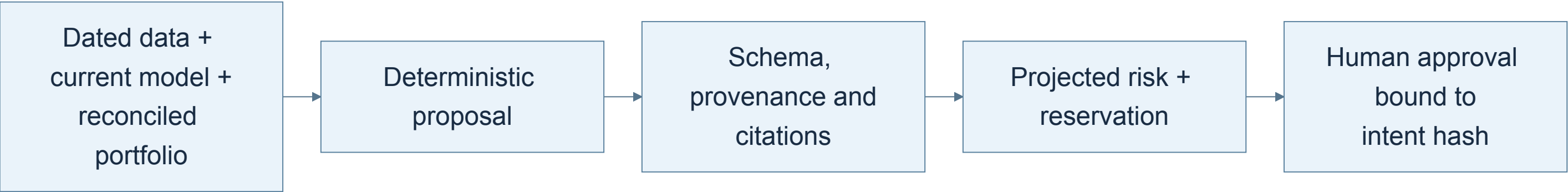
TARGET ARCHITECTURE / Six domains, one modular deployment



Control and evidence apply across every domain. LLM access is an optional adapter, not an authority layer. One Python package and one writer are the laptop baseline.

From evidence to approval

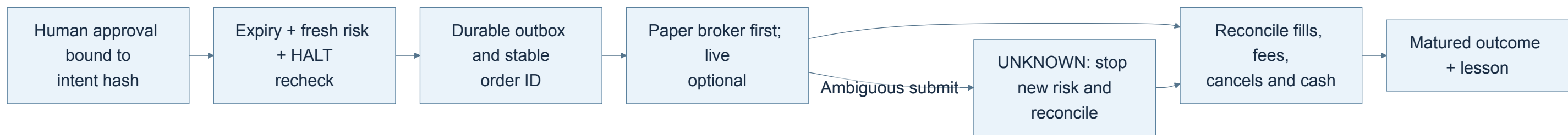
TARGET AUTHORITY FLOW / Part 1 of 2



Agents produce typed proposals, never broker orders. Approval is bound to an immutable intent; deterministic risk owns hard limits and projected reservations.

From approval to economic effect

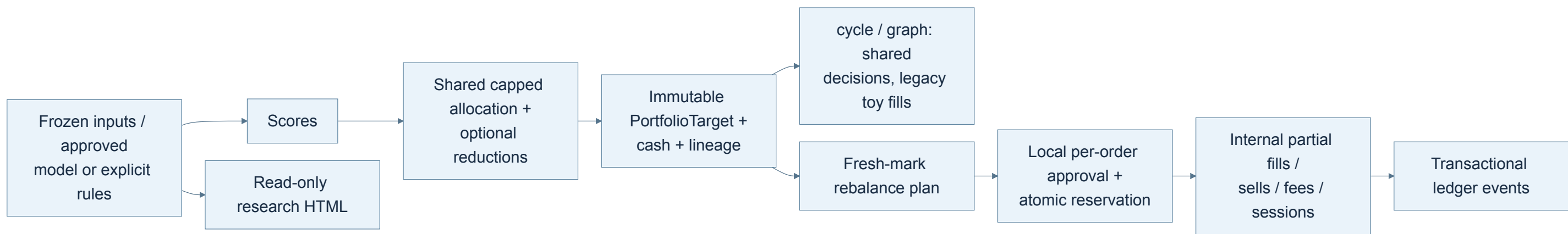
TARGET AUTHORITY FLOW / Part 2 of 2



Recheck expiry, current risk and HALT before dispatch. An ambiguous broker outcome becomes UNKNOWN and requires reconciliation, not a blind retry. These broker capabilities are not implemented in the starter.

Implemented portfolio and simulator

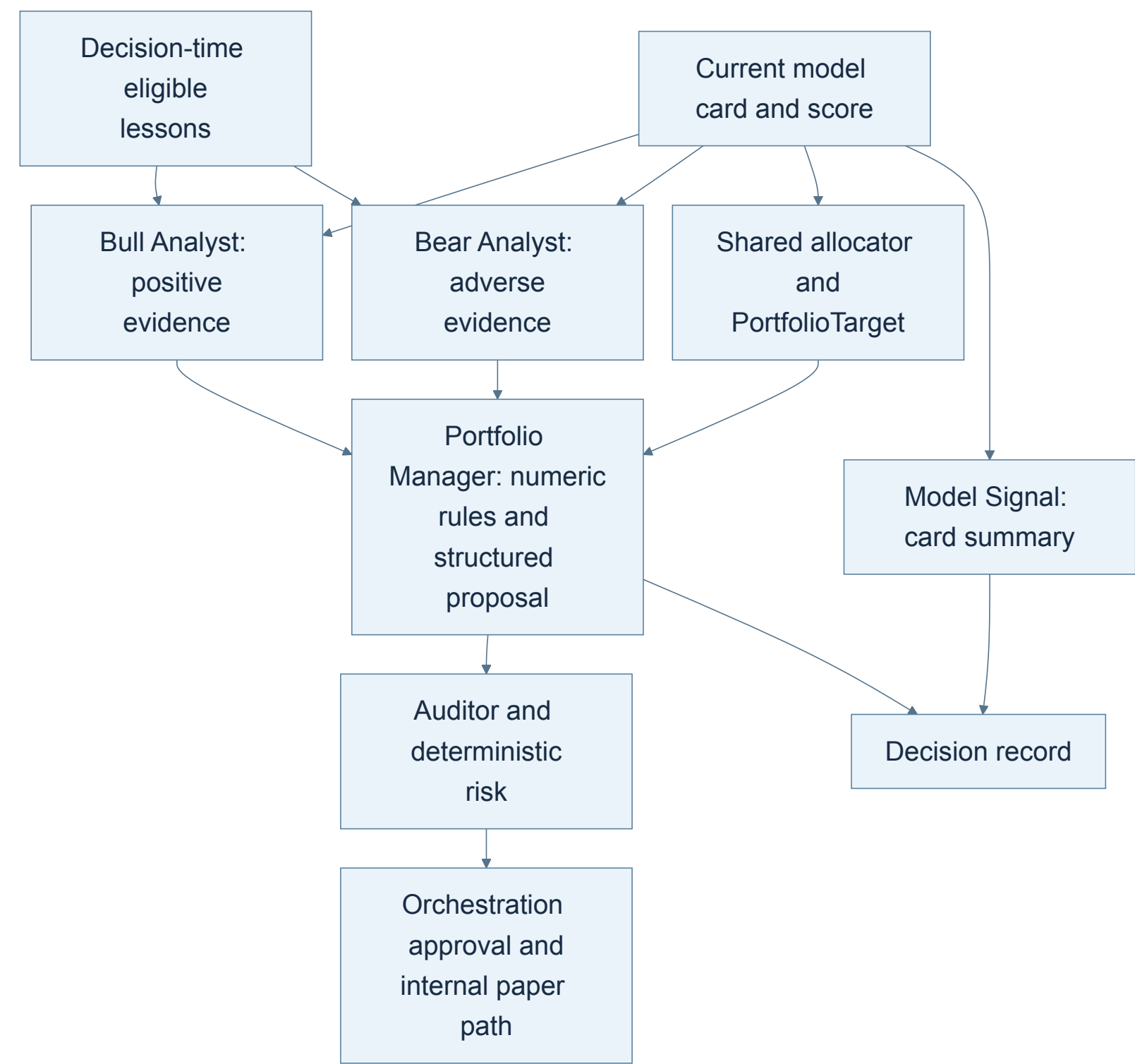
IMPLEMENTED FOUNDATIONS / Shared decisions, separate execution paths



PortfolioTarget preserves explicit cash and lineage. The durable paper CLI supports partial buys/sells, fees, marked sessions, reservations and transactional events. Legacy cycle/graph fills remain separate immediate-close demonstrations; neither is broker certification.

Four deterministic agent roles

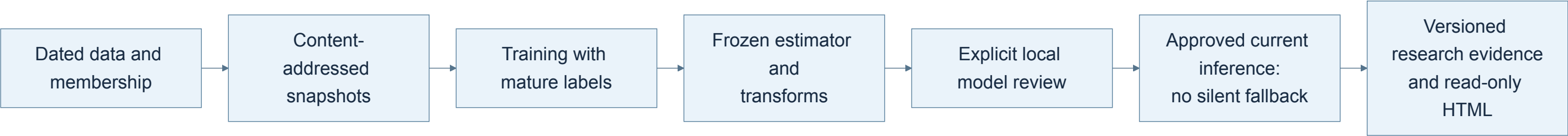
IMPLEMENTED DECISION SUPPORT / Not four autonomous LLM agents



Bull/Bear briefs do not control numeric decisions. Shared allocation supplies desired deltas; risk and trust can reduce them. Model Signal summarizes a card. Optional LLM refinement changes thesis/invalidation prose only and is disabled by default.

Frozen models and research evidence

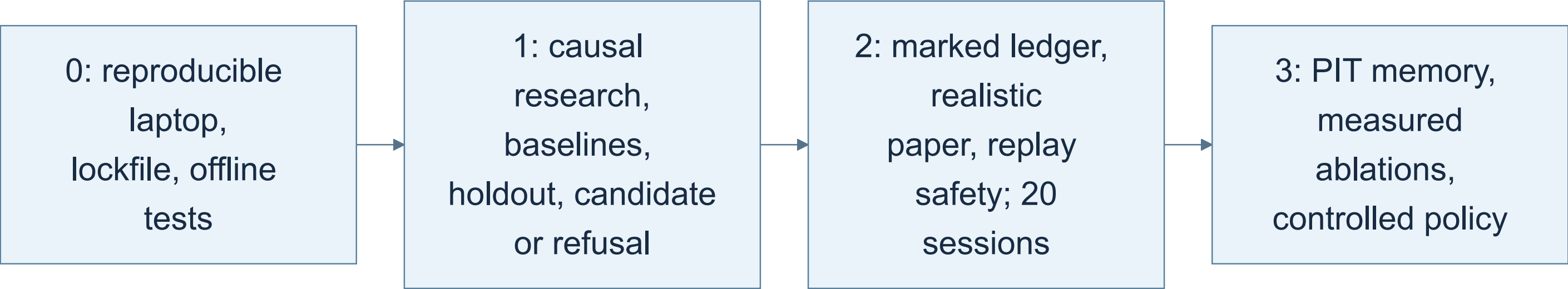
IMPLEMENTED FOUNDATIONS / Provenance before admission



Explicit rules and approved-model inference are distinct modes. Reports include benchmark costs, exposure/cash, matched-exposure baselines, fold dispersion and doubled-cost sensitivity. FinRL-X informed interfaces only: no dependency or copied execution code. Full PIT and promotion evidence remain gates.

Phases 0–3: establish evidence

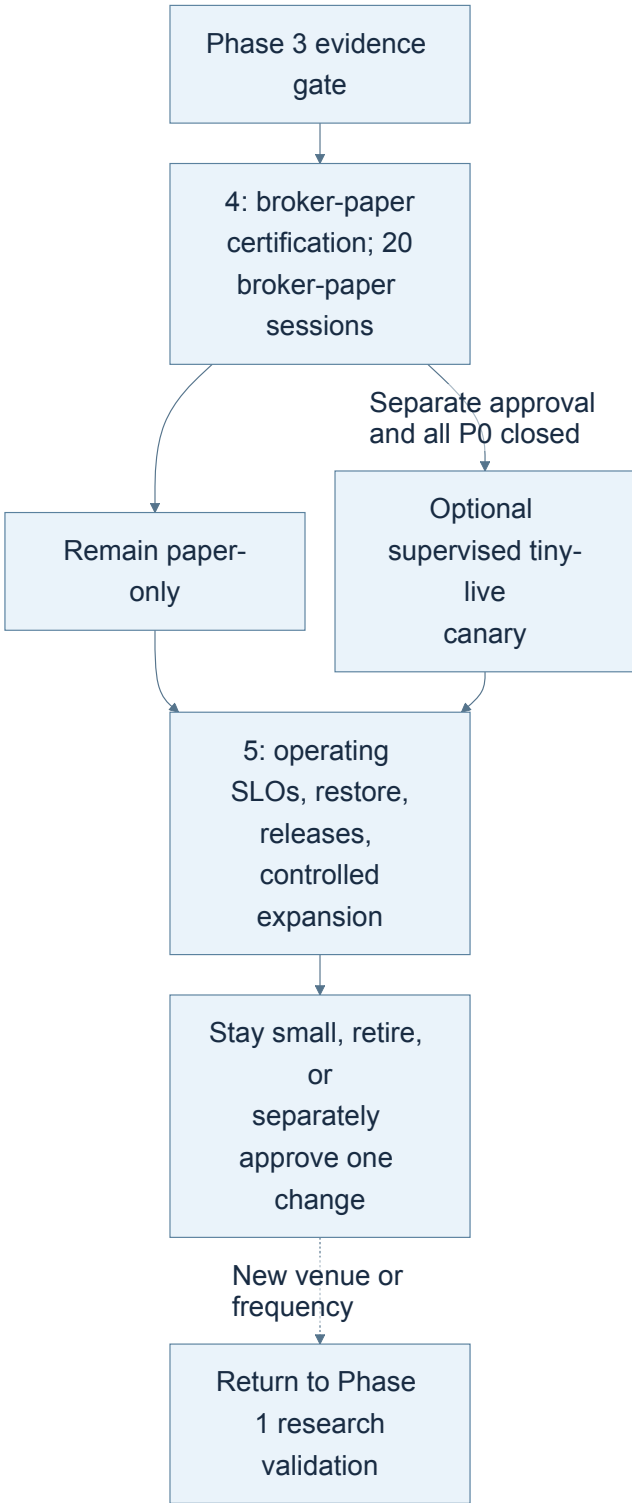
PROJECT ROADMAP / Foundations, research, paper lifecycle and measured memory



Phase 2 requires at least 20 prospective equity sessions after accounting and replay gates pass. A no-candidate research result permits simulator engineering, not live admission. Continue to Phase 4 only after the dependent evidence gates pass.

Phases 4–5: certify and operate

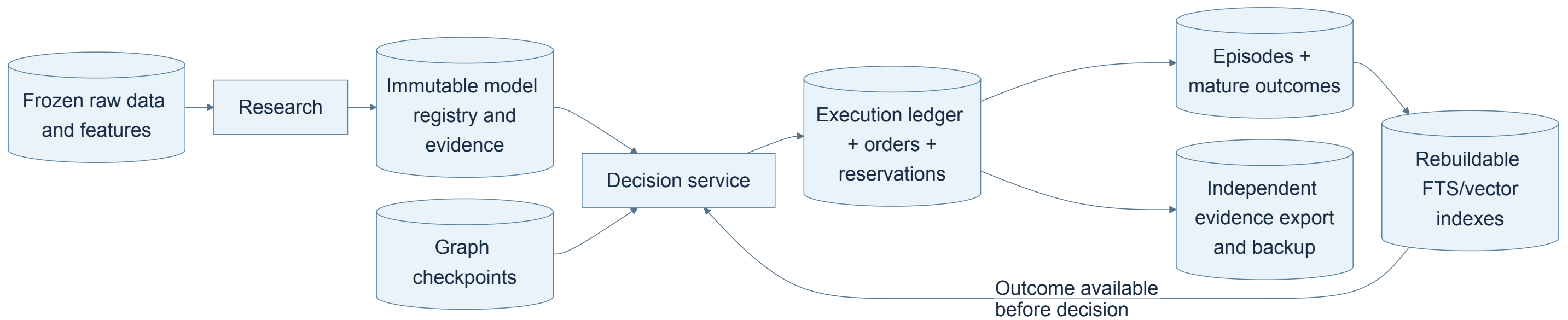
PROJECT ROADMAP / Broker paper, optional canary and governed operations



Observation windows are operating-policy floors, not proof of alpha or completion estimates. Phase 5 requires 30 calendar days of operational evidence and restore/rollback drills. P0 on the live branch means priority-zero blockers, not Phase 0.

State ownership and memory

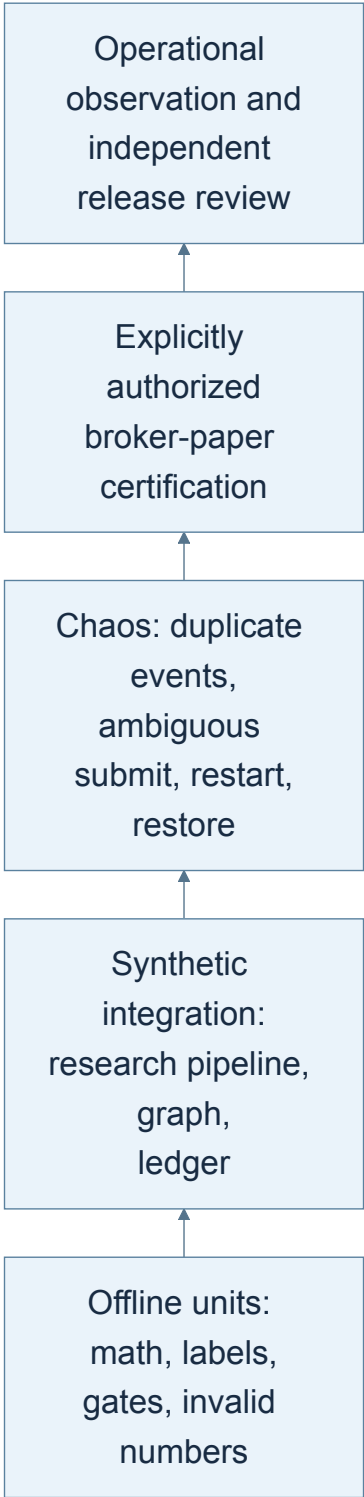
TARGET DATA RESPONSIBILITIES / Checkpoints are not portfolio truth



A restored checkpoint does not authorize a new economic effect. Preserve entry situations; retrieve only outcomes available before the decision. Indexes are rebuildable; the economic ledger is not a disposable cache.

Verification and acceptance

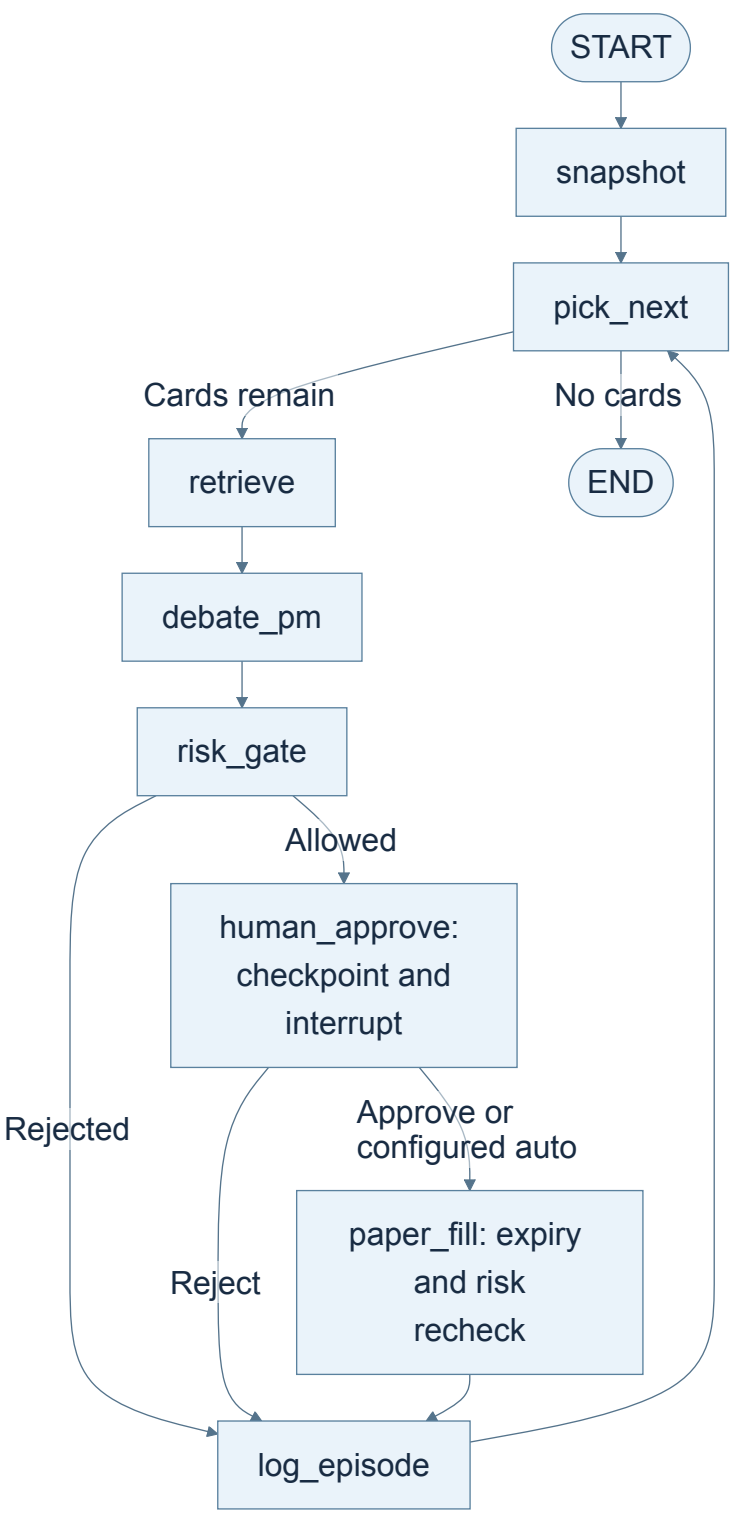
TARGET TEST PYRAMID / Evidence before external execution



Offline suites cover portfolio targets, frozen artifacts, marked ledger/reservations, decision parity, paper CLI, escaped reports and disabled shadow sentiment. They do not certify broker races, authentic PIT inputs, authorization, alpha or operational readiness.

LangGraph workflow

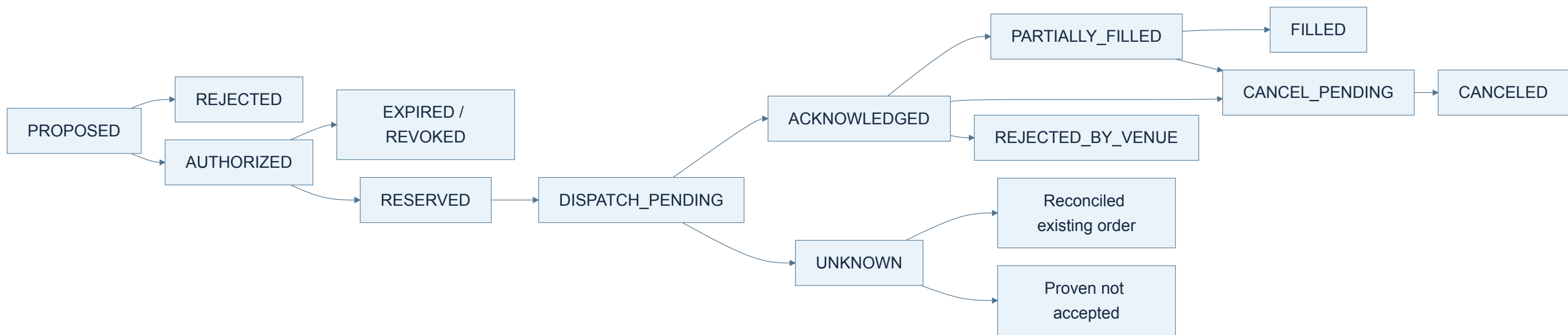
IMPLEMENTED LEGACY PATH / Shared decisions, toy fills



Graph and cycle share allocation, target construction and decision-time lesson filtering. Graph approval defaults to a 300-second snapshot expiry and rechecks risk/HALT. Its paper_fill is still a legacy toy fill, not the separate durable paper engine or a broker adapter.

Order lifecycle and uncertainty

TARGET EXECUTION CONTRACT / Not a shipped broker adapter



UNKNOWN is not rejected. Resolve acceptance through broker history and stable order IDs. A fill can race cancellation; reconcile filled quantity before releasing remaining reservations. A checkpoint is not an exactly-once transaction.

Phase outputs and admission gates

DELIVERY CONTRACT / Read each phase as an evidence gate, not a deadline

Phase	Ordered work packages	Output / admission gate
0 Reproducible laptop	0.1 tools; 0.2 locked environment; 0.3 isolated tests; 0.4 operating decisions	Clean install, green tests/lint, no accounts required, owner and budget recorded
1 Credible research	1.1 frozen data; 1.2 timing/labels; 1.3 baselines; 1.4 cost stress/holdout; 1.5 versioned model	Reproducible evidence, at least 3 valid folds where history permits, candidate or explicit refusal
2 Correct internal paper lifecycle	2.1 marked ledger; 2.2 order/exit state machine; 2.3 risk/reservations; 2.4 approval/replay; 2.5 paper observation	Accounting balances, negative tests and recovery drills; at least 20 equity sessions of prospective evidence
3 Measured memory and decision support	3.1 PIT lessons; 3.2 retrieval evaluation; 3.3 fixed/adaptive ablations; 3.4 drift/freshness; 3.5 optional LLM/policy	Frozen champion/challenger evidence; no increase in authority; disable additions without benefit
4 Broker-paper certification; optional tiny-live canary	4.1 separate adapter/account; 4.2 reconciliation; 4.3 fault injection; 4.4 broker-paper observation; 4.5 independent live decision	At least 20 broker-paper equity sessions, no unresolved safety incidents, venue evidence; optional supervised canary
5 Reliable operation and governed expansion	5.1 deploy/recover; 5.2 SLOs/budget; 5.3 release/rollback; 5.4 champion/challenger; 5.5 scope review	30 calendar days of operational evidence plus successful restore/reconciliation drill; paper-only completion is valid

Engineering completion and model admission are different. A no-candidate result allows simulator development, not live trading. Any unresolved priority-zero blocker prevents the dependent external-execution capability.